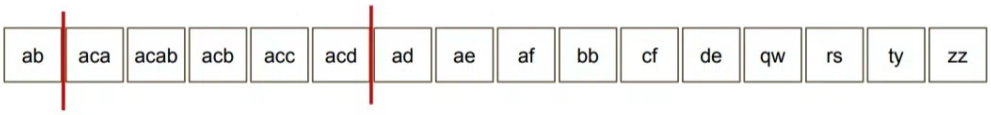
Practice problems

Binary Search ++

Given a sorted array A of strings and a query string Q, you are expected to return a list of items from A that starts with Q.

- Suggest a binary search like algorithm that returns an interval of outputs from array A.

Ex: Search for words that start with "ac" in a sorted list of words



Use binary search twice. Use one for the upper and one for lower bound

and you compare the two values and see when it changes

Clear View

Philadelphia has many tall buildings, but only some of them have a clear view of Delaware river.

- Suppose we are given an array A[1...n] that stores the height of *n* buildings on a city block, indexed from west to east.
- Building *i* has a good view of Delaware river if and only if every building to the east of *i* is shorter than *i*.

Suggest an efficient algorithm that would give a list of buildings that have a good view of Delaware river.

Make a runtime analysis of your algorithm.



Start at the end of the array checking every building keeping track of the tallest and it would have a clean view if it is the tallest so far. So compare it with the tallest.

Time: O(n)

